

JONATHAN RUEFFER

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EDUCATION

University of Maryland – College Park, MD

Expected May 2028

M.S. in Computer Science

Coursework: Computer Systems Architecture, Cryptography, Trustworthy ML, AI & Public Policy

The College of Wooster – Wooster, OH

Aug. 2022 – May 2026

B.A. in Computer Science and Mathematics, Minor in History | GPA: 3.97

Honors: *Summa cum laude*, Phi Beta Kappa, Departmental Honors in Computer Science and Mathematics, Mathematics Honor Society (Secretary), History Honor Society

Awards: Elizabeth Sidwell Wagner Prize in Mathematics, College Scholar Award, Alumni Award

Coursework: Algorithm Analysis, Data Structures & Algorithms, Deep Learning, Probability Theory, Statistical Theory, Numerical Analysis, Database Systems, Computer Networking, Programming Languages

Study Abroad: Budapest Semesters in Mathematics – Budapest, Hungary (Fall 2024)

RESEARCH EXPERIENCE

Senior Thesis (Awarded Honors) | The College of Wooster

Aug. 2025 – May 2026

Privacy-Preserving ML: An Automated, Data-Driven Approach to Parameter Selection for Homomorphic Encryption

Advisor: Prof. Alex Nord

- Developed a data-driven Python framework for CKKS parameter selection, generating and evaluating candidate configurations across security, numerical error, and runtime trade-offs.
- Evaluated search strategies across 9 dataset-model pairs and three security levels against a near-exhaustive baseline.
- Reduced average evaluation time by 55% and parameter-set count by 85% while retaining >96% of Pareto-front hypervolume.
- Presented results as a talk and poster to 50+ attendees at the Senior Research Symposium; manuscript in preparation with advisor.

Summer Research Associate | AMRE Program – The College of Wooster

May 2024 – Aug. 2024

Computer Hardware Analysis for Automated System Modeling

Advisors: Prof. Robert Kelvey, Prof. Nathan Sommer (Xavier University)

- Developed C and Python microbenchmarks to infer cache and CPU characteristics from low-level performance measurements, under a funded contract with UMD's Laboratory for Physical Sciences.
- Iteratively designed and evaluated cache-size detection methods from timing data, reducing L3 size detection time by 144× while maintaining <1 MB error.
- Implemented cache-associativity experiments using x86 cycle timing and cache-line eviction, resolving L1/L2 associativity from access-time discontinuities.
- Delivered the final technical report, client presentation, and documented codebase with a 3-student team.

TECHNICAL PROJECTS

Music Memory Journaling App | Junior Independent Study & Personal Project

Jan. 2025 – May 2025, July 2026

- Built a full-stack web app (FastAPI, React, PostgreSQL, Spotify API) that lets users attach memories to songs; automated entry-title suggestions by integrating KeyBERT keyword extraction.
- Rebuilt it as a multi-user mobile app (React Native, TypeScript) with Spotify OAuth, a tag-based Supabase schema, and row-level security isolating each user's data.

WORK EXPERIENCE

IT Intern | Nucleonova – Valencia, Spain

June 2025 – Aug. 2025

- Replaced an unmanageable Excel inventory sheet with an Android app (Kotlin) and Next.js/TypeScript web app, making 800+ items searchable for ~30 staff.
- Built multi-field search and seven composable filters over raw engineering codes with bilingual localization (EN/ES).
- Shipped features in weekly cycles, refining the interface from stakeholder feedback.

TEACHING & LEADERSHIP

Math Tutor | STEM Success Initiative – The College of Wooster

Aug. 2023 – May 2026

- Mentored ~25 Calculus I/II students per semester in class and in office hours, coordinating with professors to refine tutoring strategies and foster inclusive learning spaces.

Science Section Editor | The Voice – The College of Wooster

Jan. 2025 – May 2026

- Wrote 12+ articles on AI and science for a campus audience at a student-run newspaper ([selected articles](#)); designed print layouts (InDesign) and copyedited contributor submissions to AP style.

Teaching Assistant | Computer Science Department – The College of Wooster

Aug. 2023 – Dec. 2023

- Evaluated C++ code and proctored exams for ~40 students in Data Structures & Algorithms labs/office hours.

SKILLS

Programming Languages: Python, C++, C, TypeScript, Kotlin, JavaScript, Java, SQL, Haskell, Bash

ML & Data: PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, TenSEAL, Matplotlib

Web & Tools: Linux, Git, LaTeX, React, React Native, Next.js, FastAPI, PostgreSQL, Android SDK, Supabase, HTML/CSS

Spoken Languages: German (Fluent), Spanish (Intermediate)